

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460312.9138 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.056
Transit depth (Rp/Rs)^2	2.89 +/- 1.91 [%]
Orbital Inclination (inc)	80.38 +/- 2.85
Airmass coefficient 1 (a1)	1.001 +/- 0.052
Airmass coefficient 2 (a2)	0.002 +/- 0.074
Scatter in the residuals of the lightcurve fit is	5.11 %
Best Comparison Star	#1 - [419, 66]
Optimal Aperture	2.77
Optimal Annulus	11.84
Transit Duration (day)	0.042 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.17 ± 0.056 vs 0.1594 (deviation 0.19σ)
Duration fit vs published	0.042 d vs 0.0512 d (deviation 0.58σ)
Mid-time O–C	5.9 min
Depth SNR	3.0
Residual scatter	5.11 %

Light curve

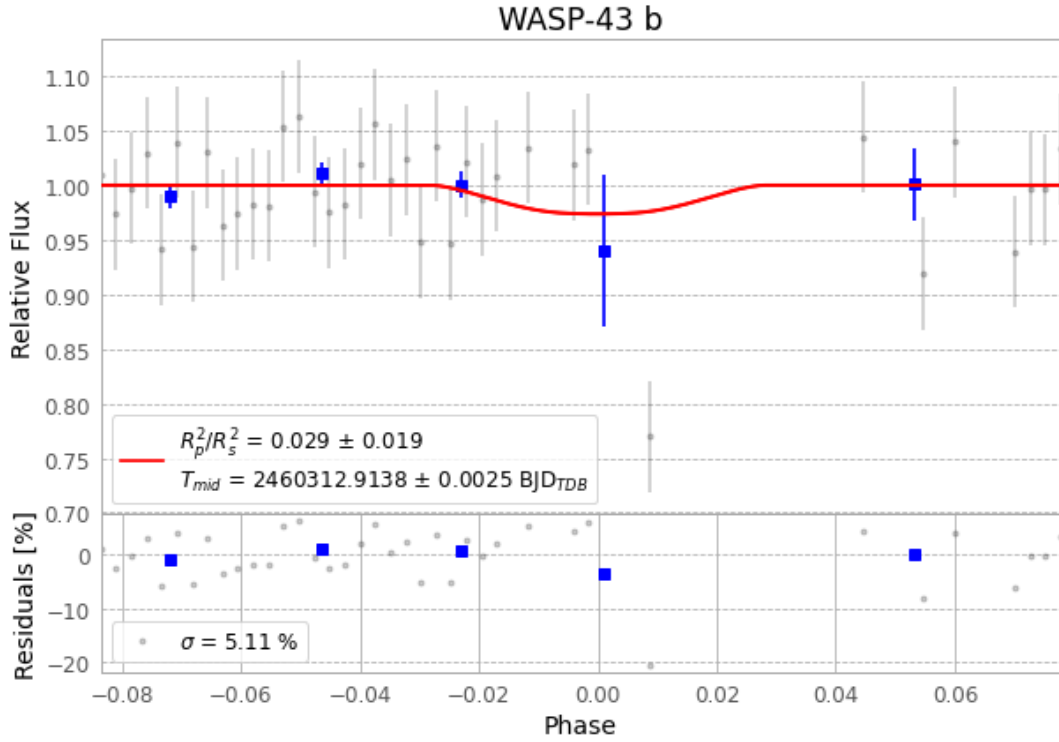


Figure 1 — FinalLightCurve_WASP-43 b_2024-01-03.png (SHA-256 1cd9a67b3a0b87b0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [496, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f0cc5fa8e961e1b7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), WASP-43240103081215.FITS → WASP-43240103112115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240103.log (81077e229c91...), FinalLightCurve_WASP-43 b_2024-01-03.png (1cd9a67b3a0b...), FinalLightCurve_WASP-43 b_2024-01-03.csv (0a47bd4e4f24...)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 01:04Z.